

Important Notes Of MySQL Part 4

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Truncate Table

TRUNCATE TABLE command is used to delete complete data from an existing table.

You can also use DROP TABLE command to delete complete table but it would remove complete table structure from the database and

you would need to re-create this table once again if you wish you

store some data.

You might choose to truncate a table instead of dropping the table and recreating it.

Truncating a table can be faster and does not affect any of the table's indexes, triggers, and dependencies.

It is also a fast way to clear out the records from a table if you don't need to worry about rolling back.

```
sql>truncate table zsalespeople;
```

Assume you have records on zsalespeople and we delete those table records which cannot be recovered back.

```
=====
=====
=====
```

=

Set Operators

union

union is a set operator which is used for combining 2 queries. The SQL UNION operator is used to combine the result sets of 2 or more SELECT statements. It removes duplicate rows between the various SELECT statements.

```
SQL> select snum from salespeople
      union
      select snum from customers;
```

SNUM

444

1001

1008

1013

1040

1089

1234

1403

1409

1432

1456

SNUM

- - - - -

1577

1666

1777

1899

1982

3453

5779

9001

9393

49494

21 rows selected.

=====

=====

union all

this operator is used for combining both the queries and both queries will get executed.

rules while using any set operator is it should have a common attribute name and data type and data type size

in the following query both the table output would come

```
SQL> select snum from salespeople
      2  union all
      3  select snum from customers;
```

SNUM

444

1001

1008

1013

1040

1089

1234

1403

1409

1432

1456

SNUM

1577

1666

1777

1899

1982

3453

5779

9001
9393
49494
1456

SNUM

1456
1001
1001
1040

26 rows selected.

SQL> select snum from customers;

SNUM

1456
1456
1001
1001
1040

```
=====
=====
=====
```

minus operator

minus operator is used for removing the common values from both the tables.

Query

write a query where u will print all salespeople who have still not been able to bring a single customer

```
SQL> select snum from salespeople
      2  minus
      3  select snum from customers;
```

SNUM

444

1008

1013

1089

1234

1403

1409

1432

1577

1666

1777

SNUM

1899

1982

3453

5779

9001

9393

49494

18 rows selected.

=====

```
=====
=====
```

```
SQL> select snum from customers;
```

```
      SNUM
-----
      1456
      1456
      1001
      1001
      1040
```

```
=====
=====
=====
```

Intersect operator

This operator which is a set operator will get printed if their is common records in both the tables.

Query

write a query where u will print
all salespeople who have booked
atleast 1 customer

```
SQL> select snum from salespeople
      2  intersect
      3  select snum from customers;
```

SNUM

1001

1040

1456

=====

=====

Joins

Joins is a facility in mysql to
combine 2 or more tables in to a

single query as per the logical requirement of the project.

You can also have 4 to 7 tables also in the same query.

There are many types of joins.

1) Equi Join : in equi join you need a common value in 2 or more tables.

Then those common values may be printed or not as per the logical requirement of the query.

query

write a query where you will print the salesman no, name and who are his customer along with their cname and

no (way to fetch all salespeople who are performing)

```
MySQL> select salespeople.snum,
sname, customers.cnum, cname
        from salespeople,
customers
        where salespeople.snum
= customers.snum;
```

	SNUM	SNAME	CNUM	CNAME
	1001	Kalia	2014	Lucy
Singh	1001	Kalia	2019	Haynes
	1456	Ranjit singh	2007	Grass
	1456	Ranjit singh	2001	Kalia
	1040	Rana Pratap		

2044

Diana

=====

=====

query

write a query where you will
print the snum, sname, and his
cnum, and cname and also print
the salesman no from sales table
(Hint use alias table names)
or

waq using equi join and alias
name print all those salsepeson
who has 1 or more customers.

```
MySQL> select s.snum, sname,  
c.cnum, c.cname, c.snum  
          from salespeople s,  
customers c  
          where s.snum =  
c.snum;
```

SNUM SNAME
CNUM CNAME
SNUM

1001 Kalia

2014 Lucy Singh

1001

1001 Kalsekhar

ia

2019

Haynes

1001

1456 Ranjit singh

2007 Grass

1456

1456 Ranjit singh

2001 Kalia

1456

1040 Rana Pratap

2044 Diana

```
1040
1013 Dr. Batli Wala
      2891 Janaki R
1013
9001 James Singh
      2828 Suganya Gowda
9001
```

7 rows selected.

=====

=====

Query

Write a query where you will
print snum, name and cnum and
cname and print only
those salesperson where customer
and salespersons reside in the
same city.

```
mySQL> select s.snum, sname,
s.city, c.cnum, c.cname, c.city
```

```

            from salespeople s,
customers c
            where s.snum =
c.snum
            and
rtrim(s.city)=rtrim(c.city);

```

```

            SNUM  SNAME
            CITY
CNUM
-----
-----
-----
CNAME                      CITY
-----
-----
            1040 Rana Pratap
            Los Angeles
2044
Diana                      Los Angeles

```

1013 Dr. Batli Wala
Mumbai
2891
Janaki R Mumbai

9001 James Singh
Mumbai
2828
Suganya Gowda Mumbai

=====
=====

Write a join where you will print
the onum, oamount and who
brought the order, and his/her
name and
who gave the order and their
names.(equi join only with 3
tables)

mySQL> select o.onum, oamount,

```
c.cnum, cname, s.snum, sname
      from salespeople s,
customers c, orders o
      where o.snum =
s.snum
      and
      o.cnum = c.cnum;
```

=====

=====

Inner Joins

Inner joins are also known as
equi joins.

The Inner Join keyword selects
all rows from both tables as long
as there is a match
between the columns. If there are
rows in the "Customers" table
that do not have matches in
"Orders", these customers will
NOT be listed
or displayed.

query

Write a query where using inner join print cnum from order table , cname from its master table, onum, amount and (print only those customers who placed a orders.)

```
mysql>select o.cnum, c.cname,  
onum, oamount  
      from customers c  
      inner join orders o  
      on c.cnum = o.cnum;
```


query

write a query where using inner

join print snum, sname, and which customer they are giving service along

with customer name and their customer number and print only those salespeople who are servicing any customers.

```
mySQL> select s.snum, s.sname,  
             c.cnum, c.cname  
           from salespeople s  
           inner join  
customers c  
           on s.snum  
= c.snum;
```

SNUM	SNAME
CNUM	CNAME
-----	-----
-----	-----
-----	-----
1001	Kalia
2014	Lucy Singh

1001 Kalia
2019 Haynes
1456 Ranjit singh
2007 Grass
1456 Ranjit singh
2001 Kalia
1040 Rana Pratap
2044 Diana
1013 Dr. Batli Wala
2891 Janaki R
9001 James Singh
2828 Suganya Gowda

7 rows selected.

=====
=====

LEFT JOIN

The LEFT JOIN keyword returns all rows from the left table (table1), with the matching rows in the right table

The result is NULL in the right side when there is no match.

query

write a query using left join
print all cname, their cnum from
orders table if they have placed
orders and also

the onum and oamount.

or

(waq to get all customers who
gave business & did not
give business.)

```
mysql> select c.cname, o.cnum,  
o.onum, o.oamount  
from customers c  
left join orders o  
on c.cnum = o.cnum;
```

CNAME

CNUM

ONUM

OAMOUNT

```

-----
-----
Suganya Gowda
Lucy Singh                2014
    3067        6543.34
Lucy Singh                2014
    3029        9494.33
Kalia
Janaki R
Haynes
Grass                      2007
    3002        87366
Grass                      2007
    3004    234564.45
Grass                      2007
    3024    44425.44
Diana                      2044
    3007        425425
Diana                      2044
    3078    87345.33

```

11 rows selected.

The above query will also print customers who have not placed orders by giving null value their.

```
=====
=====
=====
```

RIGHT JOIN

The RIGHT JOIN keyword returns all rows from the right table (table2), with the matching rows in the left table (table1). The result is NULL in the left side table when there is no match.

(this join should be used whenever we have missing records from the master table)

query

write a query where u will print
all customers name and the orders
number they have placed orders
use right join.

```
mysql> select customers.cname,  
orders.onum  
        from customers  
       right join orders  
       on customers.cnum =  
orders.cnum;
```

CNAME	ONUM
-----	-----
Grass	3024
Grass	3002
Grass	3004
Lucy Singh	3029
Lucy Singh	3067
Diana	3078
Diana	3007

7 rows selected.

```
=====
=====
=====
```

(full outer join will not work in MySQL)

```
*****
*****
```

FULL OUTER JOIN (full outer join will not work in MySQL but this query can be executed in SQL Server or Oracle SQL)

The FULL OUTER JOIN keyword returns all rows from the left table (table1) and from the right table (table2).

The FULL OUTER JOIN keyword combines the result of both LEFT and RIGHT joins.

Query

Write a query where you will print the cname, cnum from customer table and onum, cnum and order amount from order tables using full outer joins

```
mySQL> select c.cname, c.cnum,  
o.cnum, o.onum, oamount  
        from customers c  
        full outer join orders o  
        on c.cnum = o.cnum;
```

CNAME	CNUM	ONUM	OAMOUNT
Grass	2007	3004	234564.45
Diana	2044	3007	425425
Grass	2007		

2007	3002	87366
Lucy Singh		2014
2014	3067	6543.34
Lucy Singh		2014
2014	3029	9494.33
Grass		2007
2007	3024	44425.44
Diana		2044
2044	3078	87345.33
Janaki R		2891
Haynes		2019
Kalia		2001
Suganya Gowda		2828

The FULL OUTER JOIN keyword returns all the rows from the left table (Customers), and all the rows from the right table (Orders).

If there are rows in "Customers" that do not have matches in "Orders", or if there are rows in

"Orders" that do not have matches in "Customers", those rows will be listed as well.

```
=====
=====
=====
=====
```

Self Join

The SQL SELF JOIN is used to join a table to itself as if the table were two tables, temporarily renaming at least one table in the SQL statement.

It is used in rare cases of queries and you can learn about it in detail in MySQL.

```
=====
=====
```

=====